

Chirag Gupta

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AI engineer building multi-agent systems and agent infrastructure. GSoC '25 contributor and '26 mentor at Jenkins.

EXPERIENCE

AI Engineer, GoManzanas (US Startup)May 2025 - Present

Remote

- AEGIS, Multi-Agent Pipeline:** Detected pirated CAD software usage by analyzing telemetry and signal patterns.
- Built multi-agent pipeline for pirated CAD detection: 89% precision (94% manual baseline), 95% cost reduction (\$1.06 to \$0.05/case)
 - Built multi-backend LLM orchestration with fallback logic and stage-specific prompting
- Ace1t, Real-Time Voice AI:** Built voice-AI backend for a real-time accent training platform.
- Integrated multiple TTS models with ASR (OpenAI Whisper, FastConformer) for real-time voice processing
 - Designed push-to-talk pipeline with streaming audio, model-driven speech control, and silence handling
- Booster, AI Collaboration Platform:** Built multi-agent infrastructure for an ideation platform (FastAPI + React + PostgreSQL).
- Built multi-agent group chat with turn-based orchestration and real-time collaboration across agent personas
 - Designed containerized code execution environment and agent session lifecycle with idle cleanup and queue-based concurrency
 - Built custom MCP server, skills system with model tiering, and agent benchmarking framework

GSoC '25 Contributor | '26 Mentor, JenkinsMay 2025 - Present

- Built modular multi-agent diagnostic system (log-parser, classifier, recommender) for Jenkins build failures
- Added vector-store RAG layer, benchmarked on 20 curated questions: 95% context relevance, 3.75/5.0 overall score
- Integrated multi-backend LLM adapter (OpenAI, Gemini, Claude) with mode-switching for analysis vs recommendation

- AI Engineer, Reasonify Technology Pvt Ltd**
- Built multi-agent LLM pipelines for adaptive learning content and automated study-plan generation
 - Built automated Manim video generation pipeline (CLI + prompt standardization + component library), cutting production time from 5h to 20min
 - Implemented hybrid semantic filtering for TTS using regex rules and a fine-tuned Gemma 270M model

- Open Source**
- Merged upstream fixes, 2026: keephq/keep, datalayer/jupyter-mcp-server (stale-tool fallback and concurrent-miss coalescing), topoteretes/cognee, HKUDS/DeepTutor
 - Rago (OSL Incubator): Added 5 API backends (Cohere, Fireworks, Together, Groq, HuggingFace) for embeddings and generation. 7 merged PRs.
 - Keras: np.diff and np.log2 via the OpenVINO backend. Agno: Exa Answer method, deprecated-model migration across 7+ providers, Crawl4ai URL expansion.

AI & Data Science InternMay 2024 - Jan 2025

Mindcraft Labs | EvoAstra Ventures

- Mindcraft Labs: Automated 19+ workflows with RETELL, Voiceflow, and Make
- EvoAstra Ventures: Built fraud-detection model on 6.3M+ records, 94% fraud recall

PROJECTS

- Unplug:** Open-source runtime security for AI agents. Trained and shipped unplug-tiny, a DeBERTa prompt-injection span detector (94.4% recall at 0.5% FPR) that redacts the malicious span instead of dropping the whole document. 3-stage pipeline with taint tracking across tool calls; ships as a Python SDK on PyPI and a hosted API. unplug-ai.org · github.com/UnplugAI/Unplug
- Hive:** Local-first agent OS. Persistent agents from YAML, multi-model routing (Claude, Codex, LM Studio), agent rooms for collaboration. github.com/chiruu12/Hive
- Marshal:** Runs a fleet of headless coding agents (Claude Code, Codex, Cursor, OpenCode) in parallel, each in an isolated git worktree, with a per-provider cost ledger. CLI, MCP server, and Claude Code plugin; on PyPI as marshal-agents. One task across four backends, measured from the ledger: the cheapest correct run cost 1/115th of the most expensive. github.com/chiruu12/marshal

EDUCATION

M.Sc. (Hons.) Mathematics & B.E. (Hons.) Electronics & Instrumentation2021 - 2026

BITS Pilani, Goa Campus, Goa, India

SKILLS

- Languages:** Python, C/C++, SQL
- ML/AI:** PyTorch, Hugging Face Transformers, ONNX Runtime, vLLM, LangChain, MLflow
- Infrastructure:** Docker, Git, FastAPI, Pydantic, PostgreSQL, MCP, gRPC

ACHIEVEMENTS

Top 20, Meta Hacker Cup Open AI practice round (13,000+ entrants) · Winner, Smart India Hackathon 2025 · 4th place, ContextCon (YC hackathon)